

JONATHAN EARP

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Most of my hardware work is drones. I've built eight multirotors from 45 mm to 5 in, picking the motors, ESCs and props and doing the soldering and integration myself. At LBNL I built the Raspberry Pi camera units that run in people's kitchens, and I captained a VEX team that won State.

EDUCATION

University of Edinburgh Edinburgh, UK · 2026 – 2031
MEng Electrical and Mechanical Engineering, Year 1

The Head-Royce School Oakland, CA · Jun 2026
GPA 3.80 unweighted, 4.38 weighted. Honors Calculus, Advanced Physics, Advanced Chemistry, Data Science

PROJECTS

FPV and UAS Builds — Personal | *Betaflight, ArduPilot, ExpressLRS* 2023 – Present

- Built 8 multirotors from 45 mm to 5 in. Selected the motors, STM32-based ESCs and props, then soldered and integrated the flight controller, receiver, GPS and video system on each.
- Calibrate GPS, compass, barometer and IMU on every build, and tune PID loops over repeated test flights using flight logs.
- 400+ flight hours, including car-chase, real estate and music video shoots. All crash repairs and rebuilds done myself.

Head-Royce VEX Robotics — Team Captain Oakland, CA · 2022 – 2026

- Designed and built robot mechanisms, then tuned PID control with the programmers over repeated test runs.
- State Champion, Worlds Division Finalist, 2× Tournament Champion, 2× Excellence, 3× Build, 2× Amaze, 1× Design.

3D Reconstruction from FPV Flight Data — S.P.L.O.O.G.E | *Python, COLMAP, Rust, WebGL* Aug 2026

Automated Video Rendering Pipeline — P.E.T.E.R | *Python, OpenCV, Playwright, ffmpeg* 2026

EXPERIENCE

Lawrence Berkeley National Laboratory — Student Research Assistant Berkeley, CA · Apr 2024 – Present
StoveCam: low-cost camera sensor for an indoor air quality and childhood asthma study

- Built a Raspberry Pi 5 unit with RGB and thermal cameras that captures the cooktop every 60 s in occupied homes, then reworked the mounts, wiring and capture scripts until it ran reliably across different kitchens.

University of Pennsylvania, Singh Center for Nanotechnology — ESAP Participant Philadelphia, PA · Summer 2025

- Class 100 (ISO 5) cleanroom: photolithography, microfluidics, CdSe quantum dot synthesis, SEM and AFM imaging.
- Built an adaptive audio control device with Prof. Gyuseok Kim and graduate TAs, from first prototype to working demo.

SKILLS

Electrical & avionics: Brushless motors, STM32 ESCs, soldering, Betaflight, ArduPilot, ExpressLRS, analog and DJI video
CAD & fabrication: Onshape, Fusion 360, 3D printing, CNC, laser cutting, bandsaw, drill press, photolithography
Controls & sensors: PID tuning, IMU and gyro data, quaternion orientation, sensor calibration, flight-log analysis
Programming: Python (NumPy, OpenCV, Playwright), Rust, COLMAP, JavaScript, WebGL, Arduino, Raspberry Pi / Linux, Git

LEADERSHIP

- Medical Advocacy, President.** Ran a blood drive that brought in 120 donors, and set up a partnership with a local hospital.
- Eagle Scout, Senior Patrol Leader.** Taught a month-long robotics class at an elementary school for my Eagle project, and left the curriculum behind so someone else could run it again.